

1. Anthropic — Claude Sonnet 4.6

Anthropic is the company behind the Claude family of models. They released Claude Sonnet 4.6 in February 2026. What surprised me is that this model can handle up to 1 million tokens in a single conversation, which is huge. The training data goes up to August 2025, so it has fairly recent knowledge. It costs \$3 for every 1 million input tokens and \$15 for output. From what I read, this model is mainly used when you need an AI that can do multiple steps on its own — like opening a browser, writing code, or completing a task without being told each step. Teams that cannot afford the more expensive Opus model but still need strong performance tend to go with Sonnet 4.6.

2. OpenAI — GPT-5.4

OpenAI is probably the most well-known AI company right now, and GPT-5.4 is one of their recent releases from April 2026. It supports text, images, audio, and code all in one model. The context window is 1 million tokens, and pricing is \$2.50 per million input tokens. One thing I found interesting is that if you send the same prompt again, OpenAI charges only half the price through something called prompt caching. This model makes sense when your task involves complex reasoning or when you need it to connect with many third-party tools, since OpenAI has the widest number of integrations available.

3. Google DeepMind — Gemini 2.5 Flash

Google released Gemini 2.5 Flash in June 2025 and it became generally available shortly after. Out of all the models I researched, this one is the most affordable among the large context models — just \$0.30 per million input tokens. It can understand text, images, audio, and video. The context window is 1 million tokens and the knowledge cutoff is January 2025. The reason teams pick this model is that it is very fast (around 212 tokens per second) and handles large amounts of data at low cost. If a company is running thousands of API calls every day, this would be the practical choice.

4. Meta — Llama 4 Scout

Meta is different from the others because they release their models as open-weight, meaning anyone can download and run the model on their own servers for free. Llama 4 Scout was released in April 2025 and has a context window of 10 million tokens, which is the largest I found in my research. It uses something called a Mixture of Experts architecture, where only a portion of the model is active at any time, making it more efficient. Since it is free to use, it works well for organisations that deal with private data

and cannot send it to an external API. The trade-off is that you need your own infrastructure to run it.

5. xAI — Grok 4

xAI is Elon Musk's AI company, and Grok 4 is their flagship model released in June 2026. What makes it stand out is the context window of 2 million tokens, which is the largest among the paid hosted models I looked at. Pricing is \$2 per million input tokens. Another thing that is unique to Grok is that it has access to real-time data from X (formerly Twitter), so it can give you information about current events that other models might not know about. This is useful for news analysis, market research, or anything that needs up-to-date information along with strong reasoning.

6. DeepSeek — DeepSeek V3

DeepSeek is a Chinese AI company and their V3 model was released in December 2024. What made headlines when it came out was that it matched the quality of much more expensive Western models at a fraction of the price — just \$0.27 per million input tokens. It is also open-weight, so you can self-host it. The context window is 128K tokens, which is smaller compared to some others, but still enough for most practical use cases. If you are building something where the budget is tight but you still need strong text and code capabilities, DeepSeek V3 is the most cost-effective option I found.

7. Mistral AI — Mistral Large 3

Mistral is a French AI company and they released Mistral Large 3 in December 2025. It is a Mixture of Experts model with a 256K token context window, priced at around \$2 per million input tokens. Being a European company, Mistral is particularly focused on data privacy and GDPR compliance, which matters a lot for companies operating in Europe. It is also one of the better models for RAG (Retrieval Augmented Generation) use cases and agentic coding tasks. If your company is in Europe or handles data that cannot go outside certain regions, Mistral would be the first option to consider.

8. Alibaba — Qwen3.5-Plus

Alibaba released Qwen3.5-Plus in July 2026 as part of their ongoing Qwen series. The context window is 1 million tokens and the price is around \$0.50 per million input tokens, making it one of the cheapest options with a 1M context window. Alibaba has

been releasing models consistently, and the Qwen series is particularly strong in multilingual support, especially for Asian languages. If your application needs to handle Chinese, Japanese, Korean, or other Asian languages alongside English, this would be a better fit than most Western models that are trained mostly on English data.

9. Cohere — Command R+

Cohere is a company focused specifically on enterprise AI, and their Command R+ model released in 2025 reflects that focus. It has a 256K token context window and costs around \$2.50 per million input tokens. Unlike general-purpose models, Command R+ was designed specifically for RAG pipelines and tool use — meaning it is built to search through documents, call external tools, and return structured answers. If you are building an internal knowledge base, a customer support bot that reads from product documentation, or any application where the model needs to reliably retrieve and summarise information, Command R+ handles that better than most general models.

10. Moonshot AI — Kimi K3

Moonshot AI is a Chinese startup and Kimi K3 is their latest model, released in July 2026. The context window is 1 million tokens and the price is around \$1 per million input tokens — sitting between the budget options and the premium ones. It is a reasoning and coding focused model. I found this one interesting because it gives you the 1M context window at a mid-range price, whereas most other 1M models either cost \$3 or more or require you to self-host. For teams that want a cost-effective model for long-document tasks without the complexity of self-hosting, this is a reasonable option.

2. How transformers work below:

(a) Self-Attention Mechanism

Every word in a sentence looks at every other word and asks "how relevant are you to understanding me?" — then adjusts its own meaning based on the answers. That is how "bank" in "I went to the bank to deposit money" gets understood as financial institution and not a riverbank. It happens for all words simultaneously, not left to right.

(b) Vector Embeddings

Computers cannot read words, so every word gets converted into a list of numbers called a vector. The clever part is that similar words get similar numbers — so "king" and "queen" end up close to each other in this number space, while "king" and "aeroplane" are far apart. This is what the transformer actually processes, never the raw word itself.

(c) Feed Forward Neural Network

After self-attention figures out the relationships between words, the FFN processes each word individually and refines the understanding further. Think of self-attention as figuring out the context and the FFN as applying stored knowledge on top of that context. Research has shown that factual associations like "Paris is the capital of France" actually live inside these FFN layers — so it is essentially the model's memory.